

5. (a) **Complete the following** description of Type 2 diabetes by filling in the missing words. [3]

In Type 2 diabetes, body cells do not respond to the hormone
which is produced by the and so the level of
..... in the blood becomes too high.

(b) Researchers in Spain stated the following hypothesis.

‘Drinking coffee reduces the risk of having Type 2 diabetes, high blood pressure and obesity.’

- Scientists working for a large chain of coffee shops carried out an investigation to test this hypothesis.
- They used 2000 volunteers, 1 000 of whom drank coffee every day and the other 1 000 who never drank coffee.
- They recorded the number of volunteers from each group who had Type 2 diabetes, had high blood pressure or were obese.

The results of the investigation are shown in **Table 5.1**.

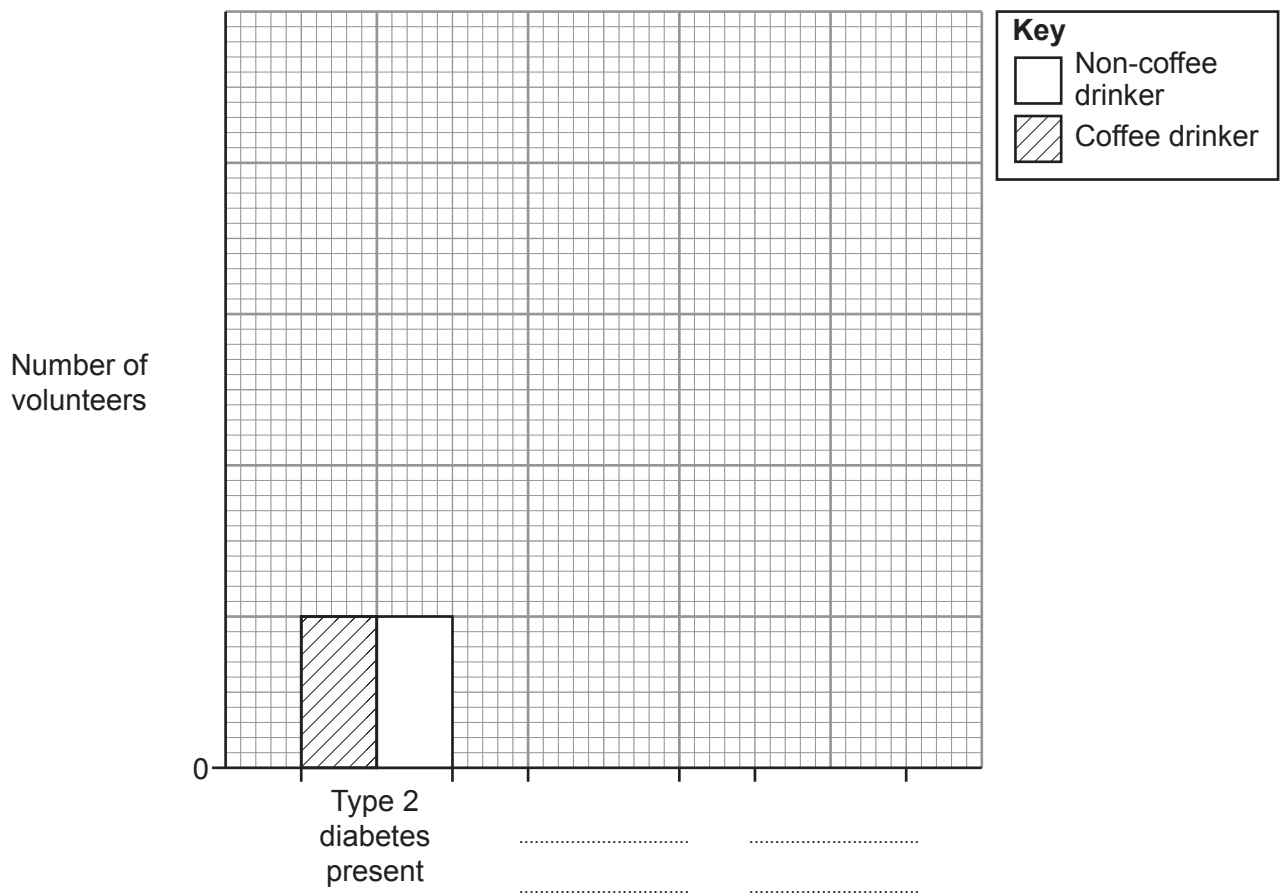
Table 5.1

Condition	Number of volunteers with the condition	
	Coffee drinkers	Non-coffee drinkers
Type 2 diabetes	100	100
High blood pressure	280	420
Obesity	340	460



- (i) Complete the bar chart in **Graph 5.2** by:
- [4]
- I. adding a scale for numbers of volunteers.
- II. drawing bars of the results for high blood pressure and obesity.
- III. labelling your bars.

Graph 5.2



- (ii) The scientists concluded that the hypothesis was only partly supported. Give the reasons for this conclusion.
- [2]

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Examiner
only

(iii) State **one** way in which the scientists should have ensured that the investigation was a fair test. [1]

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(iv) A doctor said that she did not have confidence in the results because the investigation was biased. [1]

Give **one** reason to support this point of view.

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(c) In 2019 the cost to NHS Wales of medical treatments for Type 2 diabetes was £500 million.

Suggest **one** lifestyle change which individuals can make to reduce the risk of developing Type 2 diabetes. [1]

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